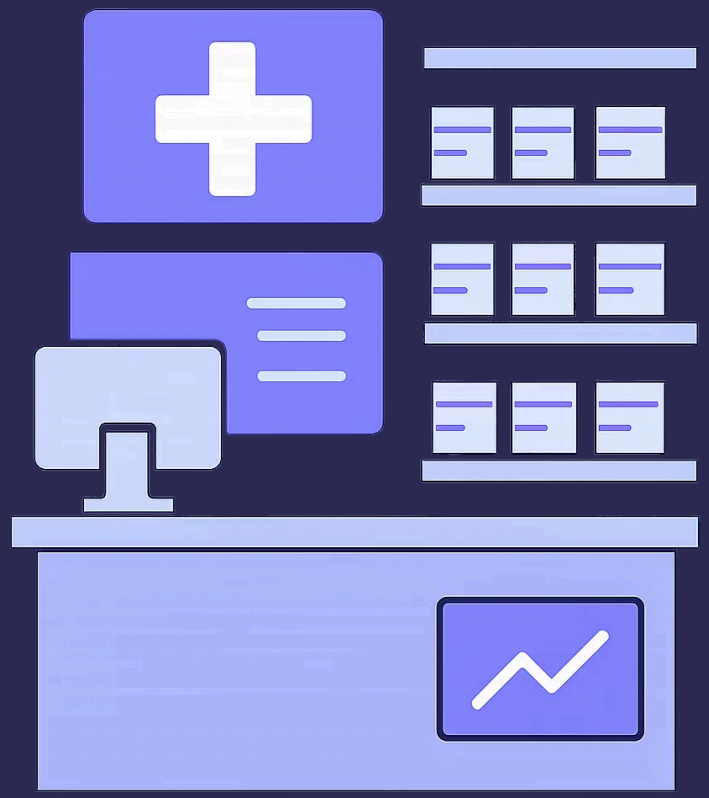

Pharma Data & Billing Solution

A Database Management System for Streamlined Pharmacy Operations

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Introduction & Problem Statement

Managing pharmacy operations manually creates inefficiencies that impact both service quality and operational costs. Errors in stock tracking, billing discrepancies, and missed expiry dates are common challenges that plague traditional pharmacy management.

Stock Errors

Manual counting leads to inventory inaccuracies and stockouts

Billing Issues

Calculation mistakes and inconsistent pricing create customer dissatisfaction

Expiry Tracking

Lost medicines due to missed expiry dates result in financial losses

Project Objectives

Efficient Medicine Management

Centralised system for adding, updating, and tracking all medicine inventory with complete product information

Automated Billing

Streamlined billing process that calculates totals, applies discounts, and generates receipts automatically

Stock Monitoring

Real-time inventory tracking with automatic updates on purchases, sales, and returns

Smart Alerts

Proactive notifications for low stock levels and approaching expiry dates to prevent losses

Key System Features

Add Medicines

Comprehensive medicine entry with name, category, price, quantity, expiry date, and supplier information

View & Search

Quick search and detailed view of any medicine with complete specifications and current stock status

Update Details

Modify medicine information including price changes, restocking quantities, and updated expiry dates

Delete Records

Remove discontinued or obsolete medicines from inventory with proper audit trail documentation

Billing Processing

Fast transaction processing with automatic calculations, receipt generation, and sales record keeping

Summary Dashboard

At-a-glance overview of key metrics including total sales, current inventory value, and alert status

Database Design

The system employs a relational database structure with normalized tables that ensure data integrity and efficient querying. Primary entities include medicines, billing records, and inventory tracking.

Core Tables

Medicines Table

MedicineID, Name, Category, Price, Quantity, ExpiryDate, Supplier, Location

Billing/Sales Table

BillID, CustomerName, Date, TotalAmount, PaymentMethod, ItemsSold

Inventory Log Table

LogID, MedicineID, TransactionType, QuantityChanged, Timestamp

System Workflow

The pharmacy management system follows a logical flow from medicine entry through to sales processing and monitoring.

Add Medicine

Enter new medicine details including name, category, pricing, and stock quantity into the database

Manage Inventory

Track stock levels, update quantities, and monitor expiry dates through the inventory management module

Process Billing

Generate sales transactions, calculate totals, and create receipts for customer purchases

Generate Alerts

System automatically flags low stock items and medicines approaching expiry for immediate attention

Dashboard Summary

View comprehensive analytics and key performance indicators on the centralised dashboard interface

Dashboard & Benefits

Time Savings

Reduces administrative tasks by up to 70%, allowing pharmacists to focus on patient care

Error Reduction

Eliminates manual calculation mistakes and data entry errors through automation

Better Tracking

Complete visibility into inventory movement, sales patterns, and expiry management

Cost Savings

Prevents losses from expired medicines and reduces overstocking through smart alerts

Conclusion & Future Scope

Project Summary

This DBMS implementation successfully addresses critical pharmacy management challenges through automated inventory tracking, billing, and intelligent alert systems. The relational database structure ensures data integrity while the intuitive interface makes daily operations efficient and error-free.

Future Enhancements

Barcode Integration

Implement barcode scanning for faster medicine entry and checkout processes

Online System

Develop web-based platform for remote access and integration with e-commerce

AI Predictions

Incorporate machine learning to forecast demand and optimise inventory levels

Thank you for your attention! We welcome your questions and feedback.